

Overview of Hardware Components

A computer consists of several hardware components that work together to process data, run applications, and store information.

1. Motherboard

The motherboard is the main circuit board of the computer. It connects all components such as the CPU, RAM, GPU, and storage devices, allowing them to communicate with each other.

2. CPU (Central Processing Unit)

The CPU is the "brain" of the computer. It performs calculations, processes instructions, and controls the overall operation of the system.

3. RAM (Random Access Memory)

RAM is temporary memory used to store data and programs currently in use. More RAM allows a computer to run multiple applications smoothly.

4. GPU (Graphics Processing Unit)

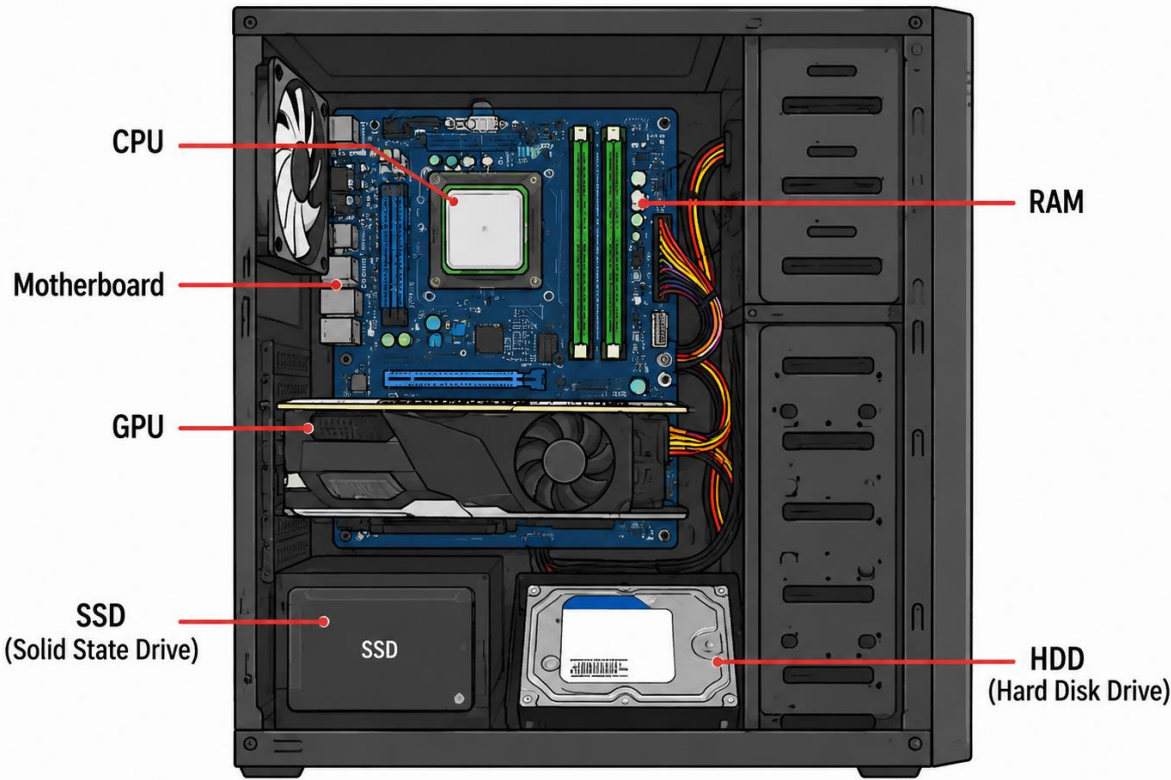
The GPU processes graphics and images. It is especially important for gaming, video editing, and 3D rendering.

5. Storage (HDD/SSD)

Storage devices permanently save the operating system, applications, and user files. Common types are HDDs and SSDs.

Task 1: Labeled Diagram of a Desktop PC

Desktop PC - Labeled Diagram



Task 2: HDD vs SSD Comparison Table

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower	Much faster
Durability	Less durable (moving parts)	More durable (no moving parts)
Noise	Makes some noise	Silent operation
Cost	Cheaper per GB	More expensive per GB
Typical Use	Large file storage, backups	Operating system, gaming, fast applications
Common Devices	Desktop PCs, storage servers	Laptops, gaming PCs, modern computers

Task 3: Role of CPU, RAM, and GPU in Gaming

CPU

The CPU processes game instructions, player actions, and game logic. In games like PUBG or Free Fire, it manages character movements, AI, and overall gameplay performance.

RAM

RAM stores game data that is currently being used. More RAM helps games load faster and prevents lag when multiple applications are running.

GPU

The GPU renders graphics, textures, and visual effects. A powerful GPU provides smoother gameplay, higher frame rates, and better image quality.

Task 4: Hardware Upgrades for Running FIFA Smoothly

Component	Why Upgrade?
GPU	Improves graphics quality and frame rates for smoother gameplay.
RAM	Allows the game and background applications to run without slowing down the system.
CPU	Handles game calculations and prevents performance bottlenecks.
SSD	Reduces game loading times and improves system responsiveness.
Motherboard (if needed)	Supports newer CPUs, RAM, and other upgraded components.

Conclusion

To run the latest FIFA smoothly, the most important upgrades are usually the **GPU**, **RAM**, and **SSD**. A balanced system ensures fast loading, smooth gameplay, and better graphics performance.